TRAUMA · NEET-SS / FET / INI-SS

Pelvis and acetabulum

1. Applied anatomy

- The pelvic ring is stable because of its **posterior ligamentous complex**: the **posterior sacroiliac ligaments (strongest ligaments in the body)**, interosseous SI ligaments, sacrospinous (resists external rotation) and sacrotuberous (resists vertical shear and flexion) ligaments, and the iliolumbar ligaments. **Posterior stability matters more than anterior.**
- Anterior: pubic symphysis (fibrocartilage); normal symphyseal width under about 5 mm, SI joint under about 4 mm.
- **Bleeding:** about 80-90% is **venous** (presacral and prevesical plexus) or from cancellous bone; about 10-15% is **arterial**, most commonly the **superior gluteal artery** (posterior, at the greater sciatic notch), then internal pudendal and obturator. The **corona mortis** (anastomosis between external iliac/inferior epigastric and obturator vessels, behind the superior pubic ramus, about 4-6 cm from the symphysis) is at risk in the Stoppa and ilioinguinal approaches.
- **Nerves:** the **L5 root** crosses the sacral ala (at risk with iliosacral screws placed too anteriorly); sacral roots in zones II/III sacral fractures; the lumbosacral trunk.

2. Pelvic ring injuries

Radiographs and CT

- AP pelvis, **inlet view** (beam angled caudally about 25-45 degrees: shows **AP displacement and rotation**) and **outlet view** (beam angled cephalad about 40-45 degrees: shows **vertical displacement** and sacral foramina). **CT** is standard.

- **L5 transverse process avulsion** suggests vertical instability. **Ischial spine avulsion** suggests sacrospinous disruption.

Young-Burgess (mechanism-based; predicts resuscitation needs)

Type	Mechanism	Injury pattern	Notes
APC I	Anteroposterior compression	Symphysis widening **under 2.5 cm**, SI ligaments intact	Stable; non-operative
APC II	AP compression	Symphysis **over 2.5 cm**, **anterior SI ligaments torn**, posterior SI intact ("open book")	Rotationally unstable, vertically stable
APC III	AP compression	Anterior and **posterior SI ligaments torn** (complete SI disruption)	Rotationally and vertically unstable; **highest blood loss and transfusion needs**
LC I	Lateral compression	Sacral buckle/impaction + oblique (horizontal) rami fractures	Stable
LC II	Lateral compression	Plus **iliac wing (crescent) fracture** or posterior SI disruption	Rotationally unstable
LC III	Lateral compression	**"Windswept pelvis"**: LC on one side, APC (open book) on the other	Unstable
VS	Vertical shear	Vertical displacement of a hemipelvis through SI joint, sacrum or ilium	Completely unstable; ipsilateral L5 transverse process avulsion
CM	Combined mechanism	Mixed patterns	

- **Lateral compression is the commonest pattern** (often elderly, side impact); **APC is associated with more bleeding**; **LC with head and chest injuries**.

Tile (stability-based)

- **A:** stable (avulsions, iliac wing, transverse sacrococcygeal).
- **B:** rotationally unstable, vertically stable (B1 open book, B2 ipsilateral LC, B3 contralateral/bucket-handle).
- **C:** rotationally **and vertically unstable** (C1 unilateral, C2 bilateral, C3 with acetabular fracture).

Sacral fractures (Denis zones)

Zone	Location	Neurological injury
I	Lateral to the foramina (ala)	About 6%; **L5 root**
II	Through the foramina	About 28%; sciatica (unilateral roots)
III	Medial to the foramina (central canal)	About 57%; **bowel, bladder and sexual dysfunction**; includes **U/H-type (spinopelvic dissociation)** fractures

Emergency management of the bleeding pelvis

1. **ATLS**, two large-bore lines, **massive transfusion protocol** (balanced 1:1:1), **tranexamic acid within 3 hours** (CRASH-2).
2. **Pelvic binder at the level of the greater trochanters** (not the iliac crests) to reduce volume in APC injuries; internally rotate and tie the legs. Avoid over-compressing LC injuries.
3. **FAST** to look for intraperitoneal bleeding.
4. **Haemodynamically unstable despite resuscitation: preperitoneal pelvic packing** (with external fixation) and/or **angioembolisation** (arterial blush on CT; embolisation is the choice for arterial bleeding); **REBOA** in selected centres. Laparotomy if FAST positive.

5. **Anterior external fixation** (iliac crest or **supra-acetabular pins**; supra-acetabular gives better control, risk to the **lateral femoral cutaneous nerve**) or a **C-clamp** for posterior instability (risk: penetration through the greater sciatic notch).

Definitive fixation

- **Symphysis: Pfannenstiel** incision, plating of the superior surface (open book APC II/III).
- **Posterior ring: iliosacral screws** (percutaneous; S1 and S2 corridors; need good inlet/outlet/lateral imaging; danger to **L5 root** anteriorly and **S1 root** through the foramen; **sacral dysmorphism** narrows the S1 corridor), anterior SI plating, posterior tension-band plating, **lumbopelvic fixation** for spinopelvic dissociation.
- **Fragility fractures of the pelvis (FFP, Rommens and Hofmann):** elderly, low-energy LC-type injuries with sacral involvement; mobilise; fix (percutaneous screws) if pain prevents mobilisation or the fracture progresses.

Associated injuries and complications

- **Urethral injury** (males, mostly **posterior (membranous-bulbar) urethra**, APC and straddle injuries): **blood at the meatus, high-riding prostate, perineal haematoma**: **retrograde urethrogram before a catheter**; suprapubic catheter.
- **Bladder rupture** (extraperitoneal commonest; repair intraperitoneal rupture surgically).
- **Open pelvic fractures** (perineal or rectal wounds): mortality up to about 50%; **diverting colostomy** for rectal/perineal wounds.
- **Morel-Lavallee lesion:** closed degloving over the greater trochanter (shearing off the fascia); infected easily; drain/debride (percutaneous or open) before or with fixation.
- **DVT/PE** (high risk; chemoprophylaxis), **neurological deficit** (L5, S1), sexual dysfunction, chronic SI pain, malunion with limb-length discrepancy and sitting imbalance.

3. Acetabular fractures

Columns and radiology

- **Anterior column:** iliac crest (anterior part) to the pubic symphysis, including the anterior wall. **Posterior column:** greater sciatic notch to the ischial tuberosity, including the posterior wall.
- **Judet views** (45-degree obliques): **obturator oblique** (shows the **anterior column** and **posterior wall**) and **iliac oblique** (shows the **posterior column** and **anterior wall**). Six radiographic lines on AP: **iliopectineal line (anterior column)**, **ilioischial line (posterior column)**, teardrop, roof, anterior rim, posterior rim.
- **Spur sign** (obturator oblique): the intact iliac bone above the joint projecting, seen in **both-column fractures** (no articular segment attached to the axial skeleton).
- **Roof arc angles (Matta):** medial, anterior and posterior roof arcs **over 45 degrees** mean the fracture spares the weight-bearing dome (non-operative may be possible); **any roof arc under 45 degrees in a displaced fracture favours surgery**. Roof arcs are **not useful for both-column and posterior wall fractures** (judge these on CT and hip stability).

Letournel classification

Elementary (5)	Associated (5)
Posterior wall (**commonest overall**)	**Both column** (commonest associated; no articular fragment attached to the ilium; spur sign)
Posterior column	T-type
Anterior wall	Transverse + posterior wall (common)
Anterior column	Posterior column + posterior wall
Transverse (only elementary type through both columns)	Anterior column/wall + posterior hemitransverse

- **Elderly:** anterior column / anterior wall patterns with **quadrilateral plate** medialisation and **superomedial dome impaction ("gull sign")** are typical (low-energy falls).

Management

- **Hip dislocation with acetabular fracture: urgent reduction** (within about 6 hours to reduce AVN), check the sciatic nerve, skeletal traction if unstable.
- **Non-operative:** undisplaced (under about 2 mm), roof arcs over 45 degrees with a congruent joint out of traction, secondary congruence of both-column fractures, small posterior wall fragments with a stable hip (**stress EUA**), elderly patients unfit for surgery.
- **Operative indications: articular displacement over 2 mm** in the dome, **unstable hip** (posterior wall over about 40-50% of the wall is usually unstable; under 20% usually stable), **incarcerated intra-articular fragments**, marginal impaction. Fix within about 2 weeks (the window closes by about 3 weeks).
- **Approaches:**

Approach	Access	Structures at risk
Kocher-Langenbeck (posterior; patient prone or lateral)	Posterior wall and column, transverse, T	**Sciatic nerve** (keep hip extended, knee flexed), **superior gluteal vessels/nerve** at the notch, **medial femoral circumflex artery** (preserve quadratus femoris and obturator externus)
Ilioinguinal (Letournel; 3 windows)	Anterior column/wall, both column	**Femoral nerve and vessels**, **lateral femoral cutaneous nerve**, **corona mortis**, spermatic cord, inguinal hernia

Approach	Access	Structures at risk
Modified Stoppa (anterior intrapelvic)	Quadrilateral plate, anterior column, both column	**Corona mortis**, obturator nerve and vessels
Extended iliofemoral	Both columns (complex late cases)	Highest **heterotopic ossification** risk

- **Elderly:** fix-and-replace (acute THA with acetabular fixation) for severe dome impaction, femoral head damage or pre-existing arthritis.

Complications

- **Post-traumatic arthritis** (commonest late complication; related to the accuracy of reduction: anatomical reduction under 1 mm gives the best results).
- **Heterotopic ossification** (posterior and extensile approaches; prophylaxis with **indomethacin** or **single-dose radiotherapy**; Brooker classification).
- **AVN of the femoral head** (with posterior dislocation), **sciatic nerve palsy** (peroneal division most affected; about 10-15%, partly iatrogenic), **DVT/PE**, infection, malreduction.

4. Hip dislocation (often with acetabular injury)

- **Posterior (about 90%):** dashboard injury, hip **flexed, adducted and internally rotated**; associated posterior wall fracture, **femoral head fracture**, **sciatic nerve injury** (peroneal division), and **ipsilateral knee injury (PCL, patella)**.
- **Anterior:** flexed, abducted and externally rotated; femoral nerve and vessels at risk.

- **Reduce urgently (within 6 hours)**; then CT for intra-articular fragments and head/wall fractures. **AVN** risk rises with delayed reduction (about 5-10% overall).
- **Thompson-Epstein** (posterior dislocation types I-V) and **Pipkin** (femoral head fractures with posterior dislocation):

Pipkin	Description	Treatment
I	Head fracture **below (caudal to) the fovea** (non-weight-bearing)	Non-operative if reduced, or excise small fragments
II	Head fracture **above (cephalad to) the fovea** (weight-bearing)	ORIF (often anterior Smith-Petersen or surgical hip dislocation)
III	I or II + **femoral neck fracture**	ORIF of neck (young) or arthroplasty (older); worst prognosis (AVN)
IV	I or II + **acetabular fracture**	Fix the acetabulum and head

Exam pearl: **APC III = most bleeding; LC = commonest; superior gluteal artery = commonest arterial bleeder; binder at the greater trochanters.** Retrograde urethrogram before catheterising a male with blood at the meatus.

Exam pearl: **Obturator oblique shows anterior column + posterior wall; iliac oblique shows posterior column + anterior wall. Spur sign = both-column fracture.** Posterior wall = commonest acetabular fracture; Kocher-Langenbeck endangers the sciatic nerve and MFCA.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.